

Question #1 of 17

Question ID: 1572847

An advantage of the bootstrap method of estimating the standard error of sample means, compared to estimating it based on a sample variance, is that the bootstrap method:

- A) only requires one sample to be taken.
- B) is less computationally demanding.
- C) can be applied to complex statistics.



Explanation

Calculating the standard error of sample means based on a single sample variance is most appropriate when the sample is unbiased and the population is approximately normally distributed. When these conditions do not hold, the bootstrap method may be more appropriate. This method is more computationally demanding in that it requires the analyst to calculate the means of multiple samples from the full dataset.

(Module 7.1, LOS 7.c)

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Question ID: 1572831

Which of the following is *least likely* a step in stratified random sampling?

- A) The population is divided into strata based on some classification scheme.
- B) The size of each sub-sample is selected to be the same across strata.
- C) The sub-samples are pooled to create the complete sample.



Explanation

In stratified random sampling we first divide the population into subgroups, called strata, based on some classification scheme. Then we randomly select a sample from each stratum and pool the results. *The size of the samples from each strata is based on the relative size of the strata relative to the population and are not necessarily the same across strata.*

(Module 7.1, LOS 7.a)

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Question ID: 1572838

An auditor who decides to handpick rather than randomly select transactions to examine for instances of fraud is *most likely* using:

- A) cluster sampling.
- B) convenience sampling.
- C) judgmental sampling.



Explanation

Judgmental sampling refers to using expert or professional judgement to select observations from a population.

(Module 7.1, LOS 7.a)

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Question ID: 1572839

Which of the following statements regarding the central limit theorem (CLT) is *least* accurate? The CLT:

- A) gives the variance of the distribution of sample means as σ^2 / n , where σ^2 is the population variance and n is the sample size.
- B) holds for any population distribution, assuming a large sample size.
- C) states that for a population with mean μ and variance σ^2 , the sampling distribution of the sample means for any sample of size n will be approximately normally distributed.



Explanation

This question is asking you to select the inaccurate statement. The CLT states that for a population with mean μ and a finite variance σ^2 , the sampling distribution of the sample means becomes approximately normally distributed *as the sample size becomes large*. The other statements are accurate.

(Module 7.1, LOS 7.b)

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Question ID: 1572833

An analyst divides the population of U.S. stocks into 10 equally sized sub-samples based on market value of equity. Then he takes a random sample of 50 from each of the 10 sub-samples and pools the data to create a sample of 500. This is an example of:

- A) simple random sampling. 
- B) stratified random sampling. 
- C) systematic cross-sectional sampling. 

Explanation

In stratified random sampling we first divide the population into subgroups, called strata, based on some classification scheme. Then we randomly select a sample from each stratum and pool the results. The size of the samples from each strata is based on the relative size of the strata relative to the population. Simple random sampling is a method of selecting a sample in such a way that each item or person in the population being studied has the same (non-zero) likelihood of being included in the sample.

(Module 7.1, LOS 7.a)

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Question ID: 1572840

The central limit theorem concerns the sampling distribution of the:

- A) sample mean. 
- B) sample standard deviation. 
- C) population mean. 

Explanation

The central limit theorem tells us that for a population with a mean μ and a finite variance σ^2 , the sampling distribution of the *sample means* of all possible samples of size n will approach a normal distribution with a mean equal to μ and a variance equal to σ^2 / n as n gets large.

(Module 7.1, LOS 7.b)

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Question ID: 1572845

If the true mean of a population is 16.62, according to the central limit theorem, the mean of the distribution of sample means, for all possible sample sizes n will be:

- A) 16.62. 
- B) indeterminate for sample with $n < 30$. 
- C) $16.62 / \sqrt{n}$. 

Explanation

According to the central limit theorem, the mean of the distribution of sample means will be equal to the population mean. $n > 30$ is only required for distributions of sample means to approach normal distribution.

(Module 7.1, LOS 7.b)

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Question ID: 1572832

An equity analyst needs to select a representative sample of manufacturing stocks. Starting with the population of all publicly traded manufacturing stocks, she classifies each stock into one of the 20 industry groups that form the Index of Industrial Production for the manufacturing industry. She then selects four stocks from each industry. The sampling method the analyst is using is *best* characterized as:

A) systematic sampling.



B) random sampling.



C) stratified random sampling.



Explanation

In stratified random sampling, a researcher classifies a population into smaller groups based on one or more characteristics, takes a simple random sample from each subgroup, and pools the results.

A random sample is one where each member of the population has an equal chance of being selected.

Systematic sampling is where every n th member of the population is selected, also known as nonrandom sampling.

(Module 7.1, LOS 7.a)

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Question ID: 1572836

Stratified random sampling is most often used to preserve the distribution of risk factors when creating a portfolio to track an index of:

A) stocks.



B) corporate bonds.



C) alternative investments.



Explanation

Stratified sampling is most often used for bond portfolios.

(Module 7.1, LOS 7.a)

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Question ID: 1572837

To estimate the average time Level I CFA candidates spend preparing for the exam, an employee of ABC Investments decides to randomly survey candidates who work at ABC's offices, although he is unsure how well they represent the candidate population. This is *most likely* an example of:

- A) stratified sampling.
- B) convenience sampling.
- C) judgmental sampling.



Explanation

Convenience sampling refers to sampling an element of a population based on ease of access.

(Module 7.1, LOS 7.a)

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Question ID: 1572842

The central limit theorem states that, for any distribution, as n gets larger, the sampling distribution:

- A) approaches a normal distribution.
- B) approaches the mean.
- C) becomes larger.



Explanation

As n gets larger, the variance of the distribution of sample means is reduced, and the distribution of sample means approximates a normal distribution.

(Module 7.1, LOS 7.b)

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Question ID: 1572846

Which technique for estimating the standard error of the sample mean involves calculating multiple means from the same sample, each with one observation removed from the sample?

A) Bootstrap.



B) Sample variance.



C) Jackknife.

**Explanation**

The jackknife technique involves calculating the standard deviation of the means from samples, each of which is calculated with a different observation removed from the original sample. The bootstrap method involves drawing multiple random samples from a dataset and calculating the standard deviation of those sample means. Standard error based on the standard deviation of a single sample is estimated by dividing the sample standard deviation by the square root of the sample size.

(Module 7.1, LOS 7.c)

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Question ID: 1572843

Which of the following is *least likely* a prediction of the central limit theorem?

A) The variance of the sampling distribution of sample means will approach the population variance divided by the sample size.



B) The standard error of the sample mean will increase as the sample size increases.



C) The mean of the sampling distribution of the sample means will be equal to the population mean.

**Explanation**

The standard error of the sample mean is equal to the sample standard deviation divided by the square root of the sample size. As the sample size increases, this ratio decreases. The other two choices are predictions of the central limit theorem.

(Module 7.1, LOS 7.b)

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Question ID: 1572841

According to the Central Limit Theorem, the distribution of the sample means is approximately normal if:

- A) the sample size $n > 30$. 
- B) the standard deviation of the population is known. 
- C) the underlying population is normal. 

Explanation

The Central Limit Theorem states that if the sample size is sufficiently large (i.e. greater than 30) the sampling distribution of the sample means will be approximately normal.

(Module 7.1, LOS 7.b)

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Question ID: 1572834

Thomas Merton, a car industry analyst, wants to investigate a relationship between the types of ads used in advertising campaigns and sales to customers in certain age groups. In order to make sure he includes manufacturers of all sizes, Merton divides the industry into four size groups and draws random samples from each group. What sampling method is Merton using?

- A) Stratified random sampling. 
- B) Simple random sampling. 
- C) Cross-sectional sampling. 

Explanation

In stratified random sampling, we first divide the population into subgroups based on some relevant characteristic(s) and then make random draws from each group.

(Module 7.1, LOS 7.a)

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Question ID: 1572844

Suppose the mean debt/equity ratio of the population of all banks in the United States is 20 and the population variance is 25. A banking industry analyst uses a computer program to select a random sample of 50 banks from this population and compute the sample mean. The program repeats this exercise 1000 times and computes the sample mean each time. According to the central limit theorem, the sampling distribution of the 1000 sample means will be approximately normal if the population of bank debt/equity ratios has:

- A) a normal distribution, because the sample is random. 
- B) a Student's t -distribution, because the sample size is greater than 30. 
- C) any probability distribution. 

Explanation

The central limit theorem tells us that for a population with a mean μ and a finite variance σ^2 , the sampling distribution of the sample means of all possible samples of size n will be approximately normally distributed with a mean equal to μ and a variance equal to σ^2/n , *no matter the distribution of the population*, assuming a large sample size.

(Module 7.1, LOS 7.b)

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Question ID: 1572835

An analyst is asked to select a sample of securities from those included in a broad-based index that can be expected to have the same return as the index while preserving the key risk exposures of the index. The analyst should *most appropriately* use:

- A) stratified random sampling. 
- B) simple random sampling. 
- C) constrained random sampling. 

Explanation

Stratified random sampling is used to preserve characteristics of an underlying dataset. (Module 7.1, LOS 7.a)